

LifeCare CMS Version 2.1

Developer Design Brief

Care Intelligence & Operating Platform Framework

Purpose

This document extends the approved LifeCare CMS Version 2.0 Quality, Compliance & Care Intelligence Framework. It is intended for software developers and solution architects building the next-generation LifeCare Care Operating System (LCOS). The objective is to transform the CMS from a documentation platform into an intelligent, scalable care operations platform.

1. Design Principles

- Retain existing CMS modules; extend rather than rebuild.
- Make the Resident Acuity Engine the core of the platform.
- Link every assessment to care planning, staffing, documentation, quality, and reporting.
- Support multi-community deployment and future home care.

2. Core Care Intelligence Engine

- Resident Assessment → Acuity Score → Level of Care → Care Package → Service Catalog → Daily Tasks → Documentation → Quality Monitoring → Continuous Reassessment.
- Recommendations generated include care plan templates, monitoring frequency, staffing demand, documentation requirements, review schedule, and quality indicators.

3. Service Catalog Engine

- Create standardized service definitions linked to SOPs, competencies, care minutes, supplies, documentation templates, quality indicators, and billing eligibility.
- Every assessment outcome should automatically generate appropriate services.

4. Staffing Intelligence

- Calculate daily care minutes, required caregiver hours, shift assignments, staffing ratios, overtime risk, and coverage gaps from resident acuity.

5. SOP Integration

- Every clinical and operational task must reference the relevant LifeCare SOP, checklist, escalation pathway, and competency requirements.

6. Quality Scorecards

- Resident Quality Score (care completion, medication compliance, nutrition, hydration, mobility, engagement, risk management).
- Community Quality Dashboard (documentation completion, incident closure, medication compliance, care plan reviews, competency status).

7. AI Decision Support

- Provide non-diagnostic recommendations based on trends (falls, appetite, weight, cognition, medication adherence, hydration, behavioral changes).
- Flag residents needing reassessment or care-plan updates.

8. Executive KPI Framework

- Clinical KPIs: falls, pressure injuries, medication errors, hospital transfers, infections, weight loss.
- Operational KPIs: task completion, documentation completion, late entries, caregiver productivity, staffing utilization.
- Financial KPIs: revenue by care level, labor cost per resident-day, occupancy, service utilization.

9. Multi-Tenant Architecture

- Hierarchy: Organization → Community → Building → Floor → Unit → Resident.
- Allow community-specific branding, pricing, SOPs, forms, and staffing while sharing a common clinical platform.

10. Canonical Data Model

- Core entities: Resident, Assessment, Acuity Score, Level of Care, Care Package, Service, Task, Observation, Care Plan, Incident, Medication, Physician Order, Staff, Competency, Audit Log, KPI.

11. Integration Roadmap

- Design APIs/connectors for accounting, HR/payroll, telemedicine, laboratories, pharmacy, wearables, nurse call, family portal, home care app, and government reporting.

12. Delivery Priorities

- Phase 1: Compliance, audit trail, RBAC, acuity engine, task generation.
- Phase 2: Service catalog, staffing intelligence, SOP integration, dashboards.
- Phase 3: AI recommendations, predictive analytics, enterprise integrations.

Expected Deliverables

- Updated solution architecture.
- Data model and ERD.
- API specifications.
- Workflow diagrams.
- UI wireframes.

- Development backlog with priorities and complexity estimates.
- Testing and validation plan.

Strategic Outcome

The completed platform should function as a Care Operating System that integrates clinical care, operations, compliance, staffing, quality management, and business intelligence. It must support Assisted Living, Independent Living, Memory Care, Transitional Care, and Home Care while remaining scalable across multiple communities.